T

COURSE SLIDES

Microsoft Azure Security Engineer Associate (AZ-500)

TGS-2023039344 | Hands-On Labs | Tertiary Infotech Academy Pte Ltd | UEN: 201200696W

Version 4

ADMIN

Copyright Notice

ADMIN

About Tertiary Infotech Academy

- WSQ course provider delivering ICT and business technology training.
- UEN: 201200696W
- Training format: instructor-led, hands-on, assessment-ready.

About the Trainer

?

Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation, software engineering and cloud learning.

DELIVERS

WSQ courses on cloud fundamentals, AI, automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Digital Attendance

- Complete SSG TRAQOM digital attendance at the required checkpoints.
- Use the same registered learner identity throughout the course.
- Assessment digital attendance is separate from class attendance.
- The TRAQOM QR code and LMS submission are provided through the LMS/TMS portal.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

ADMIN

Learner Introduction

- Share your name, role and Azure security experience.
- State one identity, network or monitoring control you want to practise.
- Confirm your access to the assigned Azure subscription or sandbox.

ADMIN

Course Objectives

Identity

Labs 1-2

RBAC, PIM, Conditional Access and managed identities

Network

Labs 3-4

NSG, ASG, Firewall, WAF and Private Endpoint controls

Platform

Labs 5-6

Bastion, JIT, encryption, SQL, Storage and Key Vault

Operations

Labs 7-8

Defender for Cloud, Policy, Sentinel and playbooks

ADMIN

Briefing for Assessment

- Assessment is open book and conducted through the LMS.
- Complete WA first, then PP.
- Use lab evidence and screenshots where instructed.
- Ask for clarification before the assessment starts, not during submission.

Assessment & Funding

Written Assessment

1 hr

Open book. Complete the WA before PP.

Practical Performance

1 hr

Use the LMS and lab evidence as instructed.

Digital Attendance

Required

Take assessment attendance before starting.

Funding

SSG

Meet attendance and assessment requirements.

ASSESSMENT

Assessment Flow Reminder

1. **TRAQOM digital attendance** >
2. **Assessment digital attendance** >
3. **Sit WA then PP** >
4. **Submit answers on the LMS** >
5. **Sign Assessment Summary Record**

The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

HANDS-ON

Access the Hands-On Labs

https://github.com/tertiarycourses/TGS-2023039344---Microsoft-Azure-Security-Engineer-Associate-AZ-500-/tree/main/labs

Option A

Clone the repository

git clone https://github.com/tertiarycourses/TGS-2023039344---Microsoft-Azure-Security-Engineer-Associate-AZ-500-.git

Option B

Download the ZIP

GitHub Code menu > Download ZIP, then open the labs folder

Run

Use the browser

Follow each lab in Azure Portal using the instructor-provided subscription

Record screenshots or notes where the lab asks for evidence. If a feature is unavailable because of licensing or permission limits, document the exact blocker.

Courseware & Assessment on the LMS

- Download the courseware and submit assessment answers at https://lms-tms.tertiaryinfotech.com/.
- Use the LMS for TRAQOM QR code, assessment attendance and submission instructions.
- Keep your learner account active until all assessment records are signed.

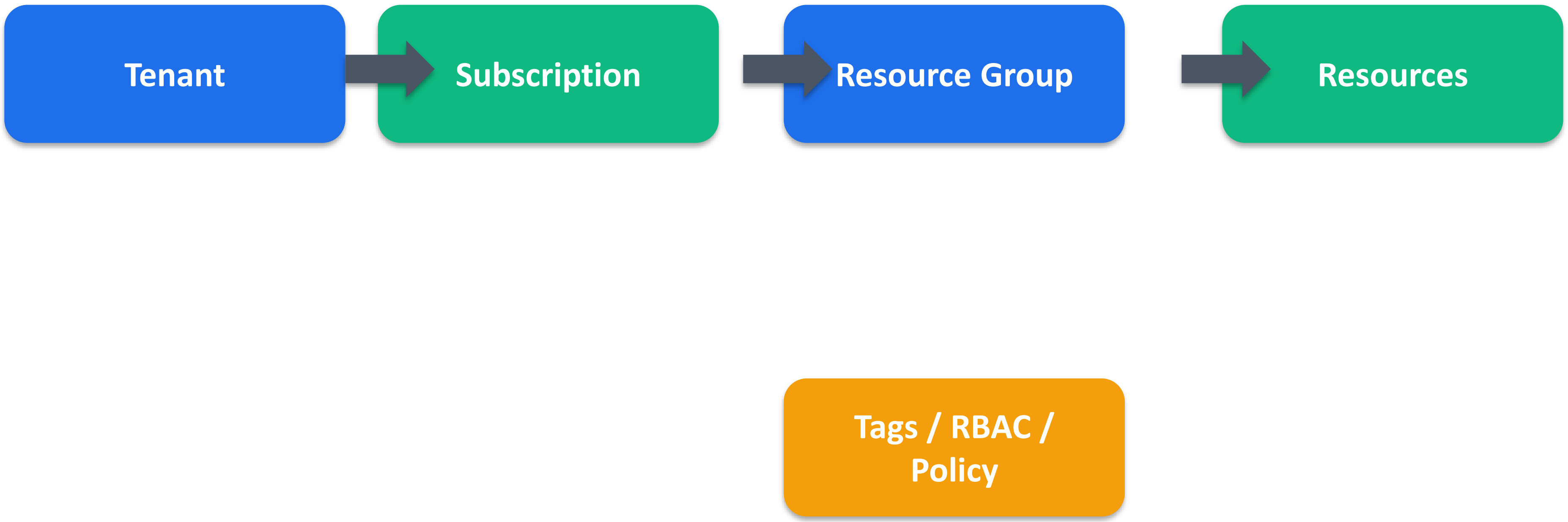
SCHEDULE

Two-Day Lesson Flow

- Day 1: Labs 1-4 - identity, privileged access, Conditional Access, managed identities and network protection.
- Day 2: Labs 5-8 - compute, data protection, Defender for Cloud, Sentinel investigation and automation.
- All lab work uses the assigned training subscription and `rg-az500-<initials>` unless your trainer instructs otherwise.

CORE DIAGRAM

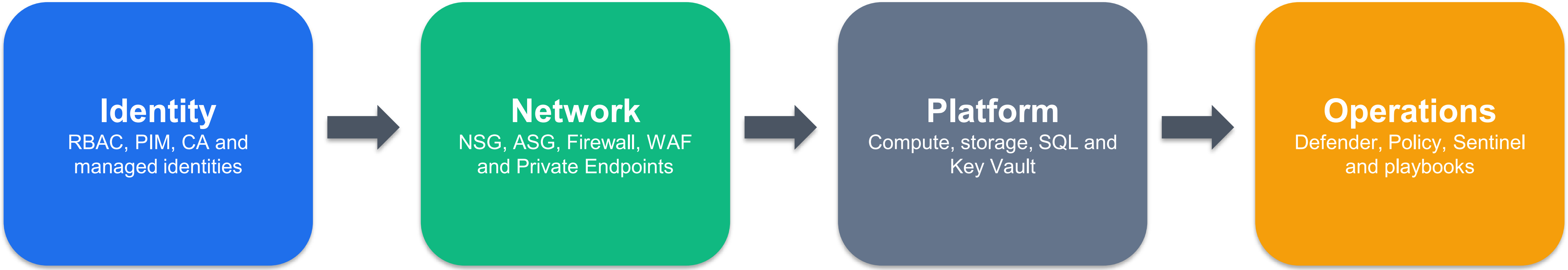
Azure Resource Model



The labs repeatedly use this hierarchy: sign in to the tenant, work inside a subscription, place resources in rg-az500-<initials>, then apply identity, network, workload, data and SecOps controls.

DIAGRAM

Two-Day Azure Lab Flow



The slide deck, Learner Guide and Lesson Plan follow this same lab sequence.

DIAGRAM

Azure Services Practised

Entra ID
RBAC, PIM, CA and app consent

→

Network
Segmentation and protected access

→

Data
Storage, SQL, secrets and keys

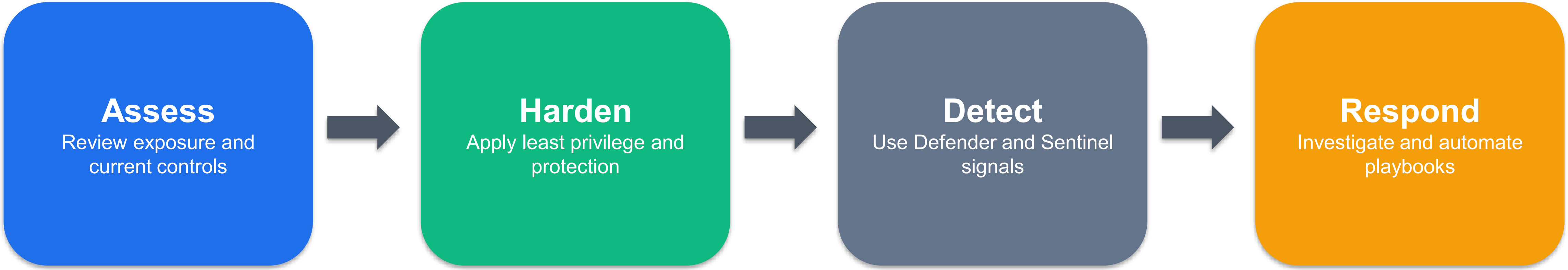
→

SecOps
Secure score, Sentinel and automation

Service groups in the diagram map directly to Labs 1-8 and the AZ-500 lab domains.

DIAGRAM

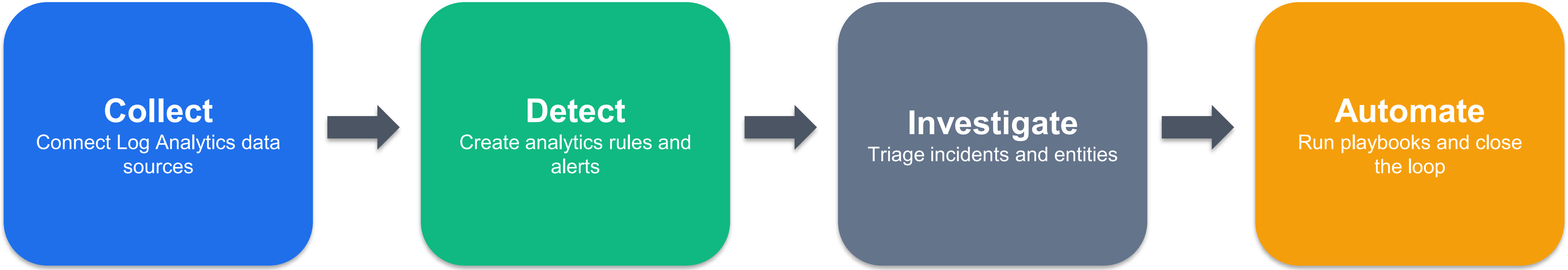
Security Posture Management Loop



The course moves from prevention to detection, investigation and response.

DIAGRAM

Sentinel Investigation Lifecycle



Lab 8 closes the course with monitoring, investigation and automation.

ALIGNMENT

AZ-500 Lab Domain Map

Domain 1

Labs 1-2

Secure identity and access

Domain 2

Labs 3-4

Secure networking

Domain 3

Labs 5-6

Secure compute, storage and databases

Domain 4

Labs 7-8

Manage security operations

DIAGRAM

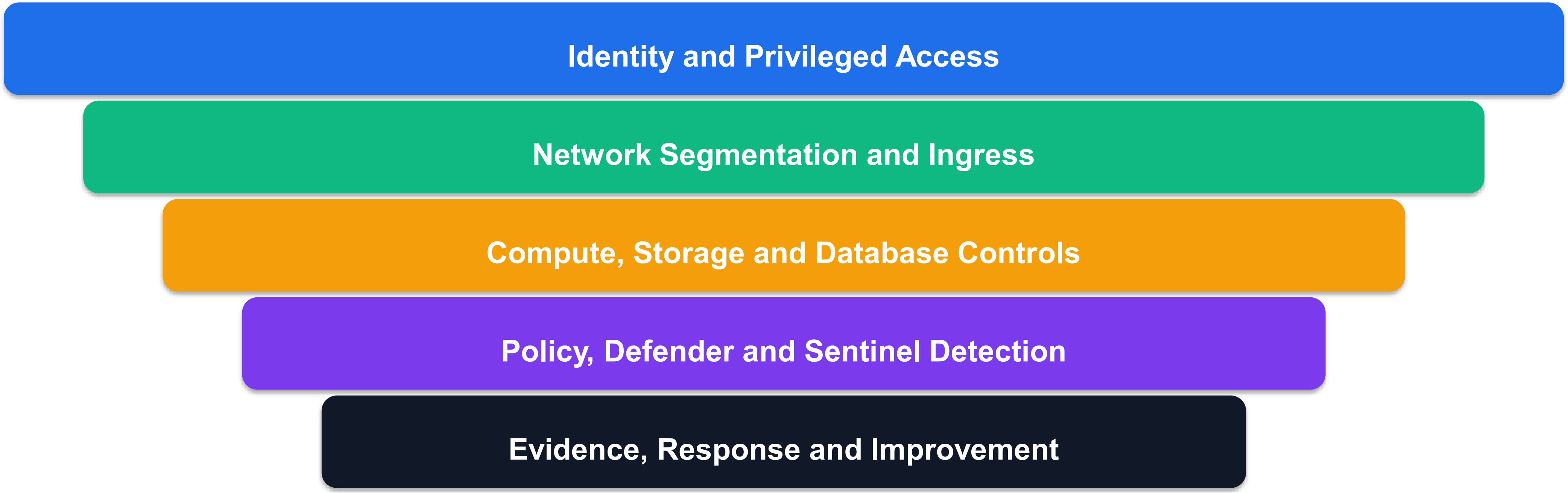
AZ-500 Lab Domain Matrix

Domain	Labs	Security Outcome	Priority
Identity and access	1-2	Least privilege and secure app access	High
Networking	3-4	Segmentation and protected ingress	High
Platform and data	5-6	Compute, storage, SQL and secrets	High
Security operations	7-8	Posture, detection and response	High

This matrix keeps the deck, Learner Guide and Lesson Plan aligned to the eight AZ-500 labs.

DIAGRAM

Azure Defense-in-Depth Layers



Learners move from preventive controls to detection, investigation, automation and evidence capture.

LAB OVERVIEW

Lab 1 - Configure RBAC, Custom Roles, and Privileged Access Review

Domain 1 - Secure identity and access

In this lab you will configure Azure role-based access control, create a scoped custom role, and review privileged access patterns with Microsoft Entra Privileged Identity Management.

Day 1

labs/lab-01-identity-rbac-pim.md

Hands-on

STEP 1

Step 1 - Create the training resource...

STEP 2

Step 2 - Review inherited access

STEP 3

Step 3 - Assign a built-in role

STEP 4

Step 4 - Create a custom role definition

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 1 - Key Steps

- Step 1 - Create the training resource group
- Step 2 - Review inherited access
- Step 3 - Assign a built-in role
- Step 4 - Create a custom role definition
- Step 5 - Review privileged identity management

CHECK

Lab 1 - Verify and Clean Up

Verification

1. How Azure RBAC scopes permissions.
2. How built-in and custom roles differ.
3. How PIM reduces standing privilege.
4. How to document missing identity licensing or permissions during security reviews.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 2 - Secure App Access with Conditional Access and Managed Identities

Domain 1 - Secure identity and access

In this lab you will review Conditional Access, app registrations, service principals, OAuth consent, and managed identities. These controls are central to secure identity and application access in Azure.

Day 1

labs/lab-02-conditional-access-managed-identities.md

Hands-on

STEP 1

Step 1 - Review Conditional Access...

STEP 2

Step 2 - Review sign-in impact

STEP 3

Step 3 - Create an app registration

STEP 4

Step 4 - Review API permissions and...

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 2 - Key Steps

- Step 1 - Review Conditional Access policies
- Step 2 - Review sign-in impact
- Step 3 - Create an app registration
- Step 4 - Review API permissions and consent
- Step 5 - Enable a managed identity

CHECK

Lab 2 - Verify and Clean Up

Verification

1. How report-only Conditional Access protects against accidental lockout.
2. How app registrations expose API permissions and consent decisions.
3. How managed identities replace application secrets.
4. How service principals connect applications to Azure RBAC.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 3 - Secure Virtual Networks with NSGs, ASGs, Peering, and Network Watcher

Domain 2 - Secure networking

In this lab you will create a segmented virtual network, apply Network Security Groups, use Application Security Groups, and validate traffic using Network Watcher.

Day 1

labs/lab-03-network-security-nsg-asg.md

Hands-on

STEP 1

Step 1 - Create a virtual network

STEP 2

Step 2 - Create application security...

STEP 3

Step 3 - Create network security groups

STEP 4

Step 4 - Associate NSGs to subnets

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 3 - Key Steps

- Step 1 - Create a virtual network
- Step 2 - Create application security groups
- Step 3 - Create network security groups
- Step 4 - Associate NSGs to subnets
- Step 5 - Use Network Watcher IP flow verify

CHECK

Lab 3 - Verify and Clean Up

Verification

1. How NSGs enforce subnet and NIC-level traffic rules.
2. How ASGs make rules easier to manage.
3. How Network Watcher validates network security behavior.
4. Why least privilege network design uses explicit tier flows.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 4 - Protect Public and Private Access with Azure Firewall, WAF, and Private Endpoints

Domain 2 - Secure networking

In this lab you will design public and private access protections using Azure Firewall, Web Application Firewall concepts, and Private Endpoints. Some resources can be reviewed instead of deployed if...

Day 1

labs/lab-04-firewall-waf-private-endpoints.md

Hands-on

STEP 1

Step 1 - Plan firewall placement

STEP 2

Step 2 - Review firewall policy rules

STEP 3

Step 3 - Review WAF protection

STEP 4

Step 4 - Create a storage account...

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 4 - Key Steps

- Step 1 - Plan firewall placement
- Step 2 - Review firewall policy rules
- Step 3 - Review WAF protection
- Step 4 - Create a storage account private endpoint
- Step 5 - Review public access settings

CHECK

Lab 4 - Verify and Clean Up

Verification

1. When to use Azure Firewall, WAF, and Private Endpoints.
2. How firewall policies organize network and application rules.
3. How private access reduces public attack surface.
4. How to document cost-sensitive security resources without deploying every component.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 5 - Secure Virtual Machines, Bastion, JIT Access, and Disk Encryption

Domain 3 - Secure compute, storage, and databases

In this lab you will secure administrative access to virtual machines and review compute hardening options such as Bastion, Just-in-Time access, and disk encryption.

Day 2

labs/lab-05-compute-security-vm-bastion-jit.md

Hands-on

STEP 1

Step 1 - Create a test virtual machine

STEP 2

Step 2 - Deploy or review Azure Bastion

STEP 3

Step 3 - Remove public management exposure

STEP 4

Step 4 - Configure Just-in-Time VM access

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 5 - Key Steps

- Step 1 - Create a test virtual machine
- Step 2 - Deploy or review Azure Bastion
- Step 3 - Remove public management exposure
- Step 4 - Configure Just-in-Time VM access
- Step 5 - Review disk encryption options

CHECK

Lab 5 - Verify and Clean Up

Verification

1. How Bastion reduces VM management exposure.
2. How JIT limits management ports to approved time windows.
3. How NSGs, Defender for Cloud, and encryption combine for VM security.
4. How to choose between platform-managed and customer-managed encryption.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 6 - Protect Storage, SQL Database, and Key Vault Secrets

Domain 3 - Secure compute, storage, and databases

In this lab you will configure security controls for Azure Storage, Azure SQL Database, and Azure Key Vault.

Day 2

labs/lab-06-storage-sql-key-vault.md

Hands-on

STEP 1

Step 1 - Harden storage account access

STEP 2

Step 2 - Review storage keys and shared...

STEP 3

Step 3 - Create a Key Vault

STEP 4

Step 4 - Add and protect a secret

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 6 - Key Steps

- Step 1 - Harden storage account access
- Step 2 - Review storage keys and shared access
- Step 3 - Create a Key Vault
- Step 4 - Add and protect a secret
- Step 5 - Create or review Azure SQL Database security

CHECK

Lab 6 - Verify and Clean Up

Verification

1. How to harden storage access and recovery settings.
2. How Key Vault protects keys, secrets, and certificates.
3. How Azure SQL Database security features map to confidentiality and audit requirements.
4. Why secret values should not be copied into lab submissions.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 7 - Manage Security Posture with Azure Policy and Defender for Cloud

Domain 4 - Manage security operations

In this lab you will use Azure Policy and Microsoft Defender for Cloud to assess compliance, secure score, and recommended remediation actions.

Day 2

labs/lab-07-defender-policy-secure-score.md

Hands-on

STEP 1

Step 1 - Assign an Azure Policy

STEP 2

Step 2 - Review compliance

STEP 3

Step 3 - Open Defender for Cloud

STEP 4

Step 4 - Review regulatory compliance

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 7 - Key Steps

- Step 1 - Assign an Azure Policy
- Step 2 - Review compliance
- Step 3 - Open Defender for Cloud
- Step 4 - Review regulatory compliance
- Step 5 - Review Defender plans

CHECK

Lab 7 - Verify and Clean Up

Verification

1. How Azure Policy enforces governance.
2. How Defender for Cloud calculates secure score.
3. How recommendations support remediation.
4. How compliance standards map technical controls to frameworks.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

LAB OVERVIEW

Lab 8 - Monitor, Investigate, and Automate with Microsoft Sentinel

Domain 4 - Manage security operations

In this lab you will configure a Microsoft Sentinel workspace, connect data, run KQL queries, create an analytics rule, and review automation options.

Day 2

labs/lab-08-sentinel-monitor-investigate-automate.md

Hands-on

STEP 1

Step 1 - Create a Log Analytics workspace

STEP 2

Step 2 - Enable Microsoft Sentinel

STEP 3

Step 3 - Connect Azure activity data

STEP 4

Step 4 - Run basic KQL queries

Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

HANDS-ON

Lab 8 - Key Steps

- Step 1 - Create a Log Analytics workspace
- Step 2 - Enable Microsoft Sentinel
- Step 3 - Connect Azure activity data
- Step 4 - Run basic KQL queries
- Step 5 - Create an analytics rule
- Step 6 - Review automation

CHECK

Lab 8 - Verify and Clean Up

Verification

1. How Sentinel uses Log Analytics workspaces.

2. How data connectors bring telemetry into the SIEM.

3. How KQL supports investigation.

4. How analytics rules and automation support alerting and response.

Cleanup Discipline

1. Run the lab cleanup section exactly.

2. Keep `rg-az500-<initials>` only when the next lab needs it.

3. Confirm no paid resource is left running.

4. Lab 8 and the trainer's final instructions cover course cleanup.

CLOSE

Course Wrap-Up

- You configured identity, network, compute, data and security operations controls across the eight labs.
- You practised evidence-based security review when licensing or permission limits blocked deployment.
- Delete remaining training resources unless the instructor asks you to preserve them for assessment.

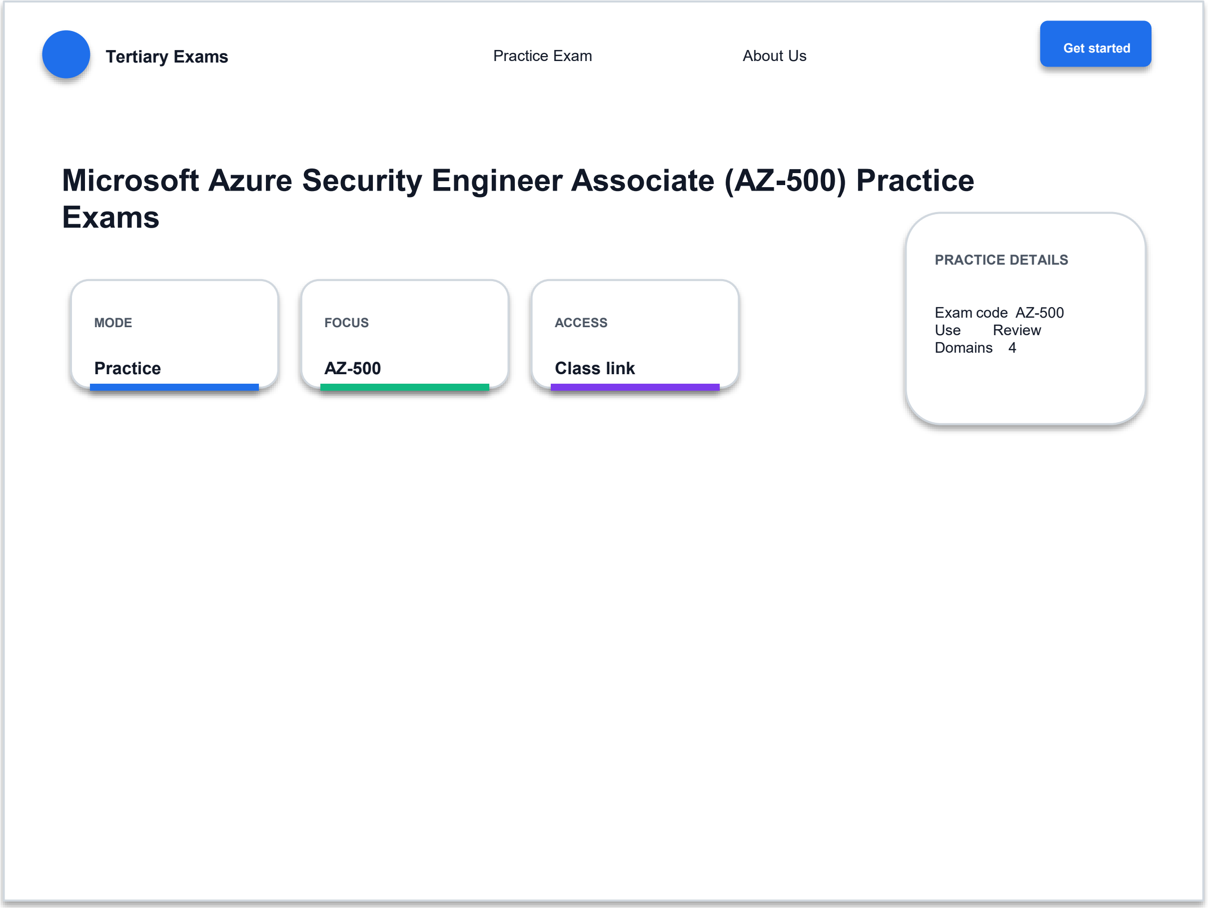
PRACTICE EXAM

Take the Online AZ-500 Practice Exams

- Reinforce AZ-500 domains with realistic, exam-style questions before assessment day.
- Use practice sets to test identity, network, platform, data and security operations controls.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to review.

Access and practise at:

Google Classroom > Practice Exams



Assessment

Written Assessment

1 hr

Open book. Complete the WA before PP.

Practical Performance

1 hr

Use the LMS and lab evidence as instructed.

Digital Attendance

Required

Take assessment attendance before starting.

Funding

SSG

Meet attendance and assessment requirements.

Assessment Flow Reminder

1. **TRAQOM digital attendance** >
2. **Assessment digital attendance** >
3. **Sit WA then PP** >
4. **Submit answers on the LMS** >
5. **Sign Assessment Summary Record**

The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Digital Attendance (Mandatory)

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- Use the same registered learner identity throughout the course.
- Assessment digital attendance is separate from class attendance.
- The TRAQOM QR code and LMS submission are provided through the LMS/TMS portal.

CLOSE

Thank You

- Submit all assessment responses on the LMS.
- Sign the Assessment Summary Record.
- Contact Tertiary Infotech Academy for post-course support.